Context-Aware Safety Assessment for Autonomous Underwater Robots in Marine Debris Management Tasks

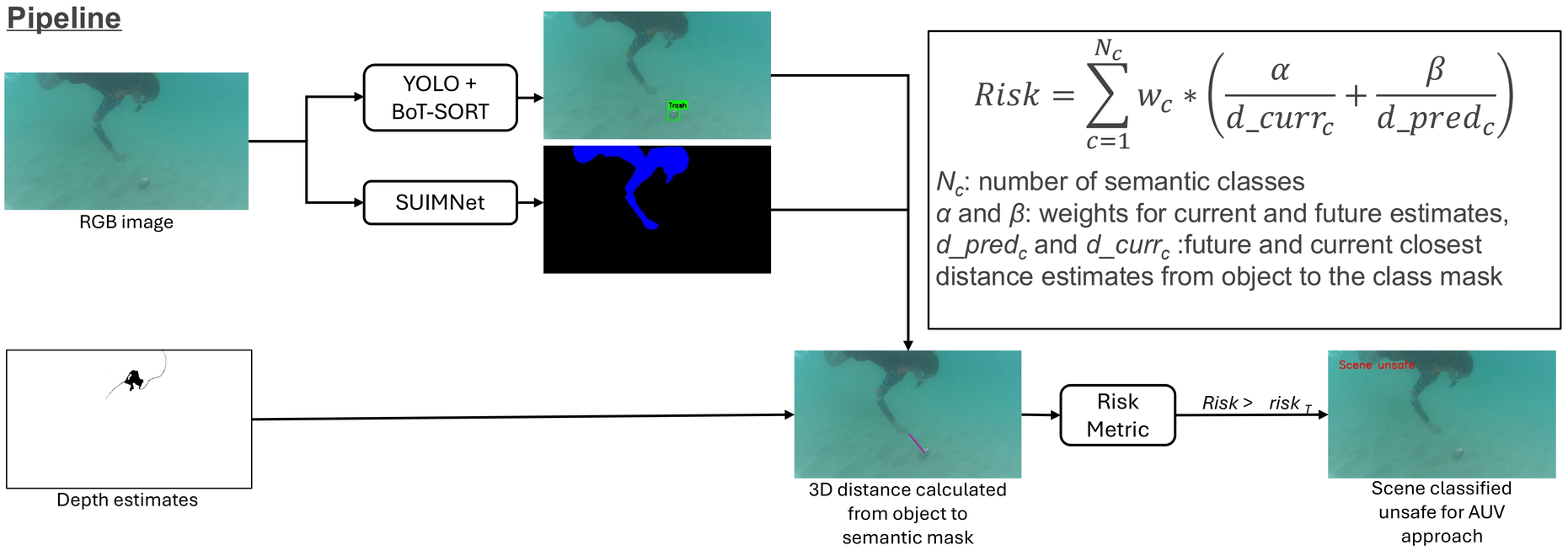
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Introduction

Marine debris removal using robots may pose a risk to sensitive underwater environments. We develop a method to consider safety and assess if a scene is suitable for robotic intervention. This approach supports safer AUV operations by reducing the likelihood of disturbing ecologically sensitive areas and improves safety in human-robot collaborative debris removal mission.

We present a pipeline to assess safety in underwater environment with respect to marine debris management, integrating debris tracking, semantic segmentation, and 3D distance.

Pipeline



$$Risk = \sum_{c=1}^{N_c} w_c * \left(\frac{\alpha}{d_curr_c} + \frac{\beta}{d_pred_c} \right)$$

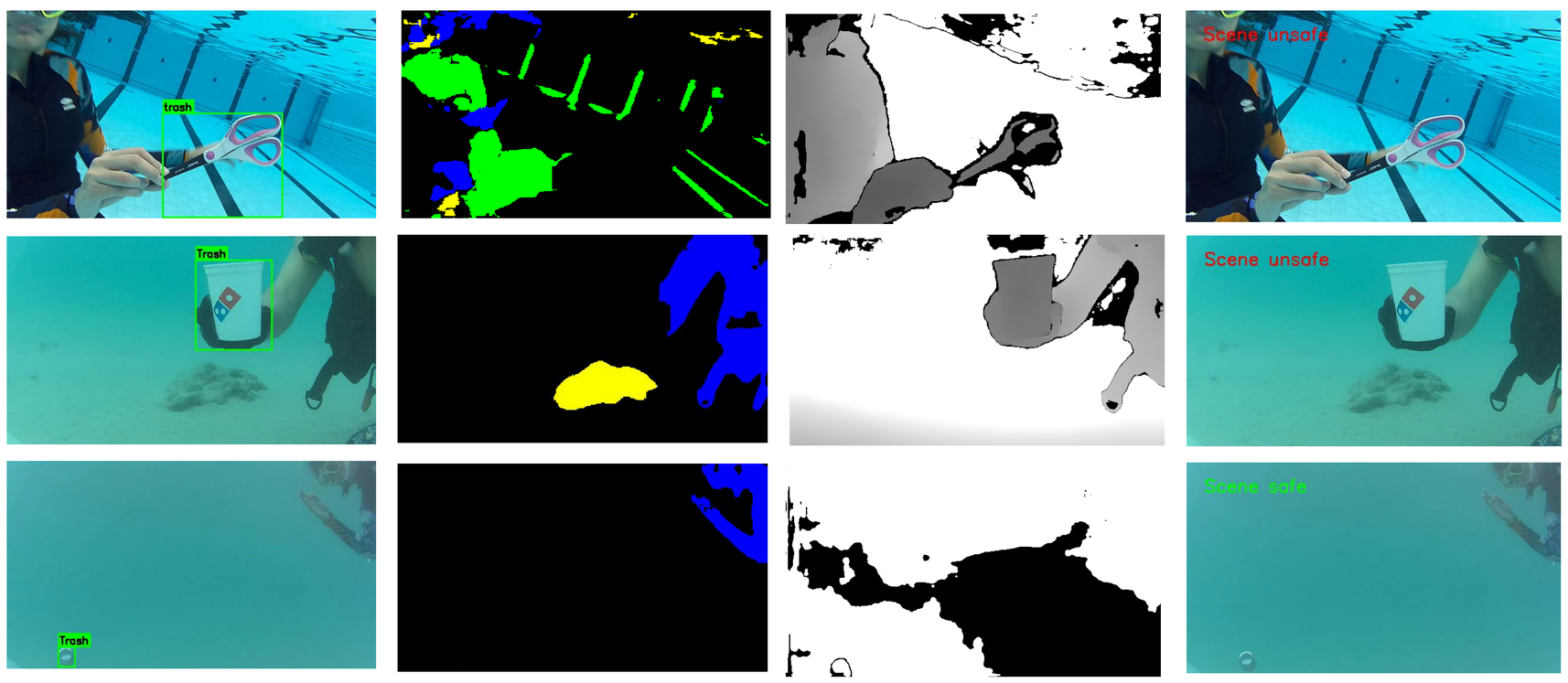
N_c: number of semantic classes
α and β: weights for current and future estimates,
d_pred_c and d_curr_c :future and current closest distance estimates from object to the class mask

Our pipeline images and depth pairs from stereo camera as our inputs. The image is fed to SUIMNet [3] to generate semantic segmentation mask and YOLO [1] with BoT-SORT [2] for debris detection and tracking. In the segmentation mask, the blue region denotes a swimmer. Using the depth data, we compute the 3D distance from the object to each semantic region and feed that to our risk metric to determine scene safety.

Sensitivity values for each semantic class. RO: Robots, FV: Fish and vertebrates, HD: Humans and divers, RI: Reef and invertebrates, WR: Wreck and ruins

Class	Weight (w_c)
RO	0.6
FV	0.2
HD	1.0
RI	0.8
WR	0.4

Results



Trash Detection | SUIM segmentation | Depth estimates | Scene classification

Qualitative results from complete pipeline deployment in pool and ocean environments. The SUIM segmentation class color scheme is: background - black, robot - red, fish and vertebrates - green, humans and divers - blue, reef and invertebrates - yellow, wrecks and ruins - magenta. The method successfully identifies scenes as safe or unsafe in these examples.

Environment	Pool	Ocean	Total
Accuracy	60.84%	77.85%	**71.31%**

2D quantitative analysis of our pipeline using YOLO and SUIMNet.

References

[1] C.-Y. Wang, I.-H. Yeh, and H.-Y. M. Liao, “YOLOv9: Learning What You Want to Learn Using Programmable Gradient Information,” 2024. [Online]. Available: https://arxiv.org/abs/2402.13616
[2] N. Aharon, R. Orfaig, and B.-Z. Bobrovsky, “BoT-SORT: Robust Associations Multi-Pedestrian Tracking,” arXiv preprint arXiv:2206.14651, 2022.
[3] M. J. Islam, C. Edge, Y. Xiao, P. Luo, M. Mehtaz, C. Morse, S. S. Enan, and J. Sattar, “Semantic Segmentation of Underwater Imagery: Dataset and Benchmark,” in 2020 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), 2020, pp. 1769–1776.

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